JOURNAL OF OPTIMIZATION THEORY AND APPLICATIONS: Vol. 97, No. 3, pp. 519-550, JUNE 1998

Optimal Ascent Trajectories and Feasibility of Next-Generation Orbital Spacecraft[1]

A. MIELE[2] AND S. MANCUSO[3]

Abstract. This paper deals with the optimization of the ascent trajectories for single-stage-to-orbit (SSTO) and two-stage-to-orbit (TSTO) rocket-powered spacecraft. The maximum payload weight problem is studied for various combinations of initial thrust-to-weight ratio, engine specific impulse, and spacecraft structural factor. For TSTO rocket-powered spacecraft, two cases are studied: uniform structural factor and nonuniform structural factor between stages.

The main conclusions are that: the design of SSTO configurations might be comfortably feasible, marginally feasible, or unfeasible, depending on the parameter values assumed; the design of TSTO configurations is not only feasible, but the payload appears to be considerably larger than that of SSTO configurations; for the case of a nonuniform structural factor, the most attactive TSTO design appears to be a first-stage structure made of only tanks and a second-stage structure made of engines, tanks, electronics, and so on.

Improvements in engine specific impulse and spacecraft structural factor are desirable and crucial for SSTO feasibility; indeed, aerodynamic improvements do not yield significant improvements in payload weight.

For SSTO configurations, the maximum payload weight behaves almost linearly with respect to the engine specific impulse and the spacecraft structural factor. The same property holds for TSTO configurations as long as the ratio of the structural factors of Stage 2 and Stage 1 is held constant. With reference to the specific impulse/structural factor domain, this property leads to the construction of a zero-payload line separating the feasibility region (positive payload) from the unfeasibility region (negative payload).

Key Words, Flight mechanics, astrodynamics, rocket-powered spacecraft, spacecraft design, single-stage-to-orbit spacecraft, two-stage-to-orbit spacecraft, optimal trajectories, ascent trajectories.

[1]This paper is based on Refs. 1-3.

[2]A. J. Foyt Professor Emeritus of Engineering, Aerospace Sciences, and Mathematical Sciences, Aero-Astronautics Group, Rice University, Houston, Texas.

[3]Graduate Student, Aero-Astronautics Group, Rice University, Houston, Texas.

0022-3239/98/0600-0519$15.00/0 © 1998 Plenum Publishing Corporation

1. Introduction

After more than thirty years of development of multi-stage-to-orbit (MSTO) spacecraft, exemplified by the Space Shuttle and Ariane three-stage spacecraft, the natural continuation for a modern space program is the development of two-stage-to-orbit (TSTO) and then single-stage-to-orbit (SSTO) spacecraft (Refs. 1–6). The first step toward the latter goal is the development of single-stage suborbital (SSSO) rocket-powered spacecraft which must take off vertically, reach given suborbital altitude and speed, and then land horizontally; this issue has been addressed in a previous paper by the present authors (Ref. 7). An example of SSSO configuration is the Lockheed-Martin X-33 spacecraft.

This paper deals with the successive orbital step, namely the optimization of the ascent trajectories for SSTO and TSTO spacecraft. Here, the optimization criterion is the maximum payload weight. Realistic constraints are imposed on tangential acceleration, dynamic pressure, and heating rate.

The optimization is done employing the sequential gradient-restoration algorithm for optimal control problems (SGRA, Refs. 8–10), developed and perfected by the Aero-Astronautics Group of Rice University. SGRA has the major property of being a robust algorithm, and it has been employed with success to solve a wide variety of aerospace problems (Refs. 11–25), including interplanetary trajectories (Ref. 11), flight in windshear (Refs. 12–15), aerospace plane trajectories (Refs. 16–18), and aeroassisted orbital transfer (Refs. 19–25).

In Section 2, we present the system description. In Section 3, we formulate the optimization problem and give results for SSTO configurations. In Section 4, we formulate the optimization problem and give results for TSTO configurations. In Section 5, we comment on the effect of drag changes on payload weight. Section 6 contains design considerations. Section 7 contains the conclusions.

2. System Description

In this paper, two configurations, intended to achieve orbital conditions at the end of the ascent, are studied: a single-stage-to-orbit (SSTO) and a two-stage-to-orbit (TSTO) rocket-powered spacecraft. The following assumptions are employed: (A1) the flight takes place in a vertical plane over a spherical Earth; (A2) the Earth rotation is neglected; (A3) the gravitational field is central and obeys the inverse square law; (A4) the thrust is directed along the spacecraft reference line; hence, the thrust angle of attack

is the same as the aerodynamic angle of attack; (A5) the spacecraft is controlled via the angle of attack and power setting.

2.1. Differential System. With the above assumptions, the motion of the spacecraft is described by the following differential system for the altitude h, velocity V, flight path angle γ, and reference weight W (Ref. 26):

$$\dot{h} = V \sin \gamma, \tag{1a}$$

$$\dot{V} = (T/W) g_e \cos \alpha - (D/W) g_e - g \sin \gamma, \tag{1b}$$

$$\dot{\gamma} = (T g_e / WV) \sin \alpha - (L g_e / WV) + (V/r - g/V) \cos \gamma, \tag{1c}$$

$$\dot{W} = -T/I_{sp}, \tag{1d}$$

in which the dot denotes derivative with respect to the time t. Here, $0 \leq t \leq \tau$, where τ is the final time. The quantities appearing on the right-hand side of (1) are the thrust T, drag D, lift L, radial distance r, local acceleration of gravity g, sea level acceleration of gravity g_e, angle of attack α, and engine specific impulse I_{sp}.

2.2. Functional Relations. In the system (1), the radius and local acceleration of gravity depend on the altitude via the relations

$$r = r_e + h, \tag{2a}$$

$$g = \mu / r^2 = \mu / (r_e + h)^2, \tag{2b}$$

where $\mu = g_e r_e^2$ is the Earth gravitational constant. Also, the specific impulse is related to the exit velocity of the gases V_{ex} by the simple expression

$$I_{sp} = V_{ex} / g_e. \tag{2c}$$

The reference weight is the so-called sea-level weight

$$W = m g_e, \tag{2d}$$

which is the product of the instantaneous mass m and the sea-level acceleration of gravity g_e. The reference weight differs from the local weight

$$W' = m g, \tag{2e}$$

which is the product of the instantaneous mass m and the local acceleration of gravity g.

Generally speaking, the aerodynamic forces can be represented by the functional relations

$$D = D(h, V, \alpha), \tag{3a}$$

$$L = L(h, V, \alpha), \tag{3b}$$

with α the angle of attack. The powerplant performance is described by the functional relations

$$T = T(h, V, \beta), \tag{4a}$$

$$I_{sp} = I_{sp}(h, V, \beta), \tag{4b}$$

with β the power setting.

2.3. Inequality Constraints. Inspection of the system (1) in light of (2)–(4) shows that the time history of the state $h(t)$, $V(t)$, $\gamma(t)$, $W(t)$ can be computed by forward integration for given initial conditions, given controls $\alpha(t)$ and $\beta(t)$, and given final time τ. In turn, the controls are subject to the two-sided inequality contraints

$$\alpha_L \leq \alpha \leq \alpha_U, \tag{5a}$$

$$\beta_L \leq \beta \leq \beta_U, \tag{5b}$$

which must be satisfied everywhere along the interval of integration. These inequality constraints can be accounted for via the trigonometric transformations

$$\alpha = (1/2)(\alpha_L + \alpha_U) + (1/2)(\alpha_U - \alpha_L) \sin x, \tag{6a}$$

$$\beta = (1/2)(\beta_L + \beta_U) + (1/2)(\beta_U - \beta_L) \sin y, \tag{6b}$$

where x and y are regarded as auxiliary control variables, providing the values of α, β are not fixed at any point of the interval of integration.

If the values of α and β are fixed at the initial point and/or at the final point (this is the case with the particular problem under consideration), then α, β and for that matter x, y must be treated as auxiliary state variables. This is achieved via the supplementary transformations

$$\dot{x} = u, \tag{7a}$$

$$\dot{y} = w, \tag{7b}$$

or

$$\dot{x} = u, \tag{8a}$$

$$\dot{y} = w^2, \tag{8b}$$

with x, y auxiliary state variables and u, w, auxiliary control variables. The representation (7) allows α and β to increase or decrease as needed along the trajectory. The representation (8) allows α to increase or decrease as needed, but forces β to decrease monotonically along the trajectory; see the boundary conditions (19f) and (20f). We employ (8) preferentially to (7) so as to force the power setting to decrease monotonically along the trajectory.

2.4. Aerodynamic Forces. The aerodynamic forces (3) are given by

$$D=(1/2)C_D\rho(h)SV^2, \tag{9a}$$

$$L=(1/2)C_L\rho(h)SV^2, \tag{9b}$$

where C_D is the drag coefficient, C_L the lift coefficient, S a reference surface area, and ρ the air density (Ref. 27). Disregarding the dependence on the Reynolds number, the aerodynamic coefficients can be represented via polynomial functions of the angle of attack, specifically,

$$C_D=A_0(M)+A_1(M)\alpha+A_2(M)\alpha^2, \tag{10a}$$

$$C_L=B_0(M)+B_1(M)\alpha+B_2(M)\alpha^2, \tag{10b}$$

with coefficients $A_i(M)$ and $B_i(M)$ depending only on the Mach number $M=V/a(h)$, where a is the speed of sound. These coefficients are computed via least-square fit of available aerodynamic data at various Mach numbers and angles of attack (Refs. 16–18). Here, the assumptions $A_1=0$, $B_2=0$ are employed; the remaining coefficients A_0, A_2 and B_0, B_1 are shown in Figs. 1–4 versus the Mach number.

As explained, the angle of attack is subject to Ineq. (5a). For the present configuration, the bounds in (5a) are given by

$$\alpha_L=-3 \text{ deg}, \qquad \alpha_U=+3 \text{ deg}, \tag{11a}$$

hence

$$-3\leq\alpha\leq+3 \text{ deg}, \tag{11b}$$

with the implication that (6a) and (8a) become

$$\alpha=K\sin x, \qquad K=3 \text{ deg}=0.05236 \text{ rad}, \tag{12a}$$

$$\dot{x}=u, \tag{12b}$$

with x an auxiliary state variable and u an auxiliary control variable.

2.5. Powerplant Model. For the rocket powerplant under consideration, the following simplified form of Eqs. (4) is assumed:

$$T=\beta T_*, \tag{13a}$$

$$I_{sp}=I_{sp*}, \tag{13b}$$

where β is the power setting, T_* a reference thrust (thrust for $\beta=1$), and I_{sp*} a reference specific impulse. The fact that T_* and I_{sp*} are assumed to be constant means that the weak dependence of T and I_{sp} on altitude and Mach number, relevant to a precision study, is disregarded within the present feasibility study.

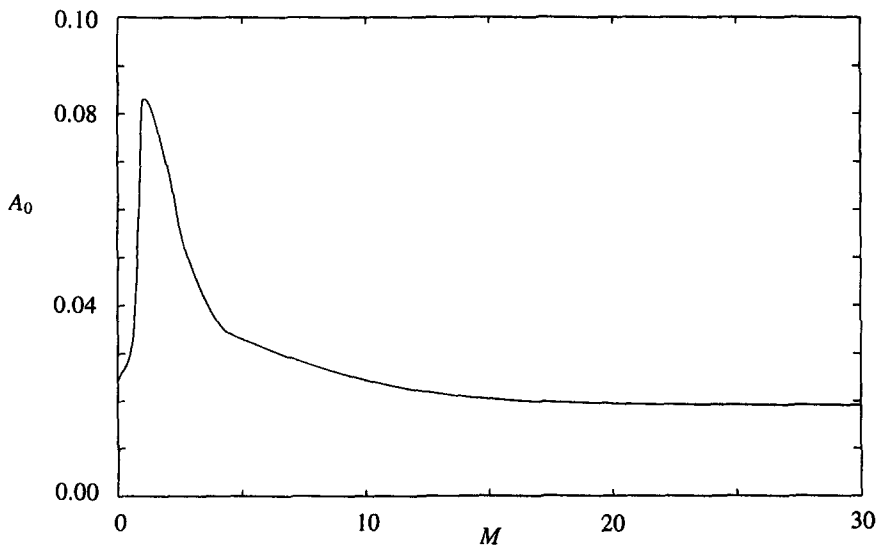

Fig. 1. Coefficient A_0 vs Mach number.

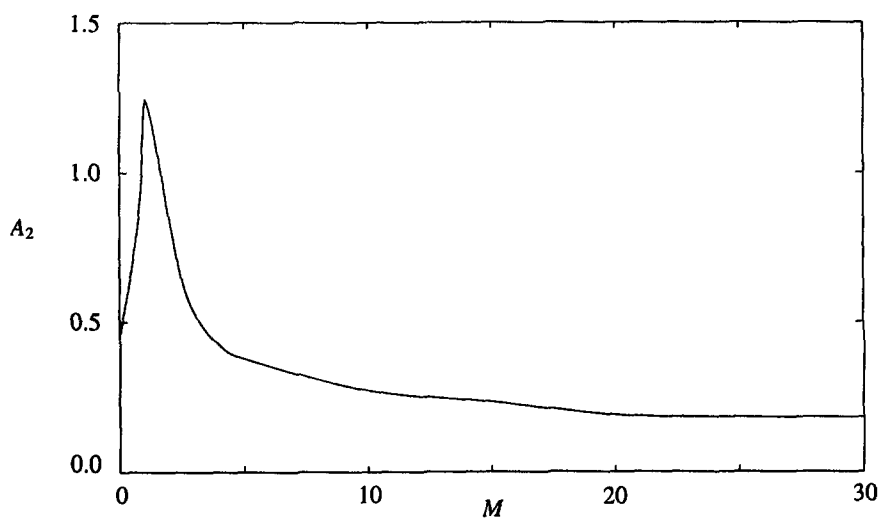

Fig. 2. Coefficient A_2 vs Mach number.

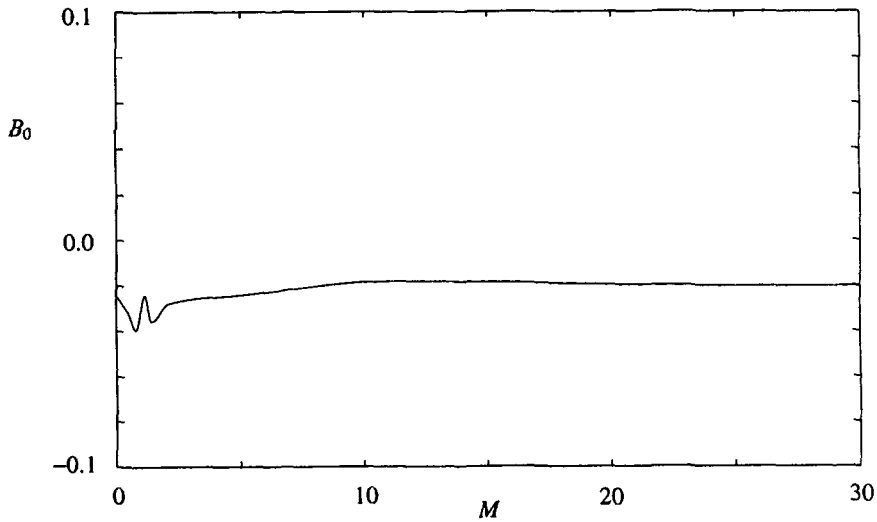

Fig. 3. Coefficient B_0 vs Mach number.

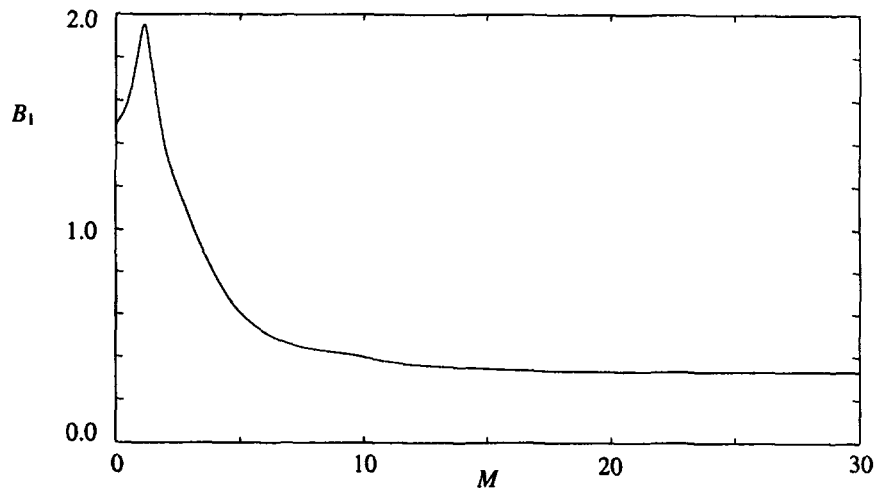

Fig. 4. Coefficient B_1 vs Mach number.

As explained, the power setting is subject to Ineq. (5b). For the present configuration, the bounds in (5b) are assumed to be

$$\beta_L = 0, \qquad \beta_U = 1, \tag{14a}$$

hence

$$0 \leq \beta \leq 1, \tag{14b}$$

implying that T_* is the same as the maximum thrust and further implying that Eqs. (6b) and (8b) become

$$\beta = (1/2)(1 + \sin y), \tag{15a}$$

$$\dot{y} = w^2, \tag{15b}$$

with y an auxiliary state variable and w an auxiliary control variable.

2.6. Path Constraints. Let the tangential acceleration a_T, dynamic pressure q, and heating rate Q per unit time and unit surface area be described by the relations

$$a_T = (T/W)g_e \cos \alpha - (D/W)g_e - g \sin \gamma, \tag{16a}$$

$$q = (1/2)\rho(h)V^2, \tag{16b}$$

$$Q = C\sqrt{[\rho(h)/\rho(h_*)]}(V/V_*)^{3.07}. \tag{16c}$$

In Eq. (16c), $h_* = 100$ kft is a reference altitude and $V_* = 10$ kft/s a reference velocity; under the assumption that the nose radius is $r_n = 1.0$ ft, the constant in (16c) has the value $C \cong 102$ BTU/ft^{2}s. In addition to the control inequality constraints (11b) and (14b), converted into state equality constraints via the transformations (12) and (15), some path constraints can be imposed on the quantities (16), for instance,

$$a_T \leq 3\, g_e, \tag{17a}$$

$$q \leq 700 \text{ lb/ft}^2, \tag{17b}$$

$$Q \leq 150 \text{ BTU/ft}^2\text{s}. \tag{17c}$$

2.7. Atmospheric Model. The atmospheric model used is the 1976 US Standard Atmosphere (Ref. 27). In this model, the values of the density are tabulated at discrete altitudes. For intermediate altitudes, the density is computed by assuming an exponential fit for the function $\rho(h)$. This is equivalent to assuming that the atmosphere behaves isothermally between any two contiguous altitudes tabulated in Ref. 27.

2.8. Supplementary Data. The following data have been used in the numerical experiments:

$$\text{radius of Earth } r_e = 20925 \text{ kft}, \tag{18a}$$

$$\text{Earth gravitational constant } \mu = 0.14076 \text{ E}+08 \text{ kft}^3/\text{s}^2, \tag{18b}$$

$$\text{sea-level acceleration of gravity } g_e = 32.174 \text{ ft/s}^2, \tag{18c}$$

$$\text{spacecraft initial wing loading } W_i/S = 387 \text{ lb/ft}^2. \tag{18d}$$

The value (18d) is perhaps close to that of the Lockheed-Martin Venture Star SSTO spacecraft.

3. SSTO Configuration

For the SSTO spacecraft, the maximum payload weight problem is studied for initial thrust-to-weight ratio σ in the range 1.3 to 1.5, engine specific impulse I_{sp} in the range 400 to 500 s, and structural factor ϵ in the range 0.08 to 0.12.

3.1. Boundary Conditions. The initial conditions ($t = 0$, subscript i) are

$$h_i = 0 \text{ kft} = 0 \text{ km}, \tag{19a}$$

$$V_i = 0 \text{ kft/s} = 0 \text{ km/s}, \tag{19b}$$

$$\gamma_i = \pi/2, \tag{19c}$$

$$W_i/W_r = 1, \tag{19d}$$

$$\alpha_i = 0 \Rightarrow x_i = 0, \tag{19e}$$

$$\beta_i = 1 \Rightarrow y_i = \pi/2. \tag{19f}$$

The final conditions ($t = \tau =$ free, subscript f) are

$$h_f = 1519 \text{ kft} = 463 \text{ km}, \tag{20a}$$

$$V_f = 25.04 \text{ kft/s} = 7.632 \text{ km/s}, \tag{20b}$$

$$\gamma_f = 0, \tag{20c}$$

$$W_f/W_r = \text{free}, \tag{20d}$$

$$\alpha_f = \text{free} \Rightarrow x_f = \text{free}, \tag{20e}$$

$$\beta_f = 0 \Rightarrow y_f = 3\pi/2. \tag{20f}$$

In Eqs. (19)–(20), the reference weight W_r is the same as the initial weight W_i. The velocity (20b) is the circular velocity at the altitude (20a).

3.2. Weight Distribution. Once the final weight W_f is known for a given initial weight W_i, one can compute the propellant weight W_p, structural weight W_s, and payload weight W_* using the following relations:

$$W_i = W_f + W_p, \tag{21a}$$

$$W_f = W_s + W_*, \tag{21b}$$

$$\epsilon W_p = (1-\epsilon) W_s, \tag{21c}$$

where

$$\epsilon = W_s/(W_s + W_p) \tag{22}$$

denotes the spacecraft structural factor. These relations admit the solutions

$$W_p = W_i - W_f, \tag{23a}$$

$$W_s = \epsilon(W_i - W_f)/(1-\epsilon), \tag{23b}$$

$$W_* = (W_f - \epsilon W_i)/(1-\epsilon), \tag{23c}$$

which in normalized form become

$$W_p/W_i = 1 - W_f/W_i, \tag{24a}$$

$$W_s/W_i = \epsilon(1 - W_f/W_i)/(1-\epsilon), \tag{24b}$$

$$W_*/W_i = (W_f/W_i - \epsilon)/(1-\epsilon). \tag{24c}$$

3.3. Optimization Problem. For the SSTO configuration, the maximum payload problem can be formulated as follows:

$$\max \quad J = W_* = (W_f - \epsilon W_i)/(1-\epsilon), \tag{25a}$$

$$\text{s.t.} \quad (1), (12), (15), (17), (19), (20). \tag{25b}$$

The unknowns are the functions

$$h(t),\ V(t),\ \gamma(t),\ W(t),\ x(t),\ y(t),\ u(t),\ w(t),\ \tau, \tag{26}$$

which include the original state variables h, V, γ, W, the auxiliary state variables x, y, the auxiliary control variables u, w, and the parameter τ.

With the solution known, the original control variables α, β can be recovered via (12a) and (15a). The following comments are pertinent.

(i) Concerning the path constraints (17), the nature of the problem is such that (17b) and (17c) are satisfied always with strict inequality. Therefore, only (17a) is accounted for via a penalty functional added to the functional being minimized.

(ii) In treating problem (25), it is convenient to normalize the dimensional time t via the transformation

$$\theta = t/\tau, \tag{27a}$$

where θ is the dimensionless time and τ the final time. Clearly,

$$0 \leq \theta \leq 1, \qquad \text{if } 0 \leq t \leq \tau. \tag{27b}$$

(iii) At the initial point, a special singularity is present in Eq. (1c), due to the fact that $V_i = 0$. For this reason, we split the trajectory into two phases. Phase 1 is flown with constraints

$$\gamma = \pi/2, \tag{28a}$$

$$\beta = 1. \tag{28b}$$

In this phase, the angle of attack is small so that

$$\sin \alpha \cong \alpha. \tag{28c}$$

In this way, the system (1) becomes a 3rd order system, the singularity is avoided, and α is uniquely determined in Phase 1. This phase is terminated when the velocity reaches the value 0.3 kft/s. At this point, Phase 2 is started and the complete system (1) is treated as a 4th order system. Typically, the time length of Phase 1 is on the order of 1/50 of the time length of Phase 2.

(iv) The optimization problem under consideration is of the Bolza type and can be solved via the sequential gradient-restoration algorithm for optimal control problems (SGRA, Refs. 8–10). Since SGRA is a gradient-type algorithm, it can handle singular optimal control problems, while this is not the case with second-order algorithms.

3.4. Results. For the SSTO configuration, a parametric study with respect to the parameters σ (initial thrust-to-weight ratio), I_{sp} (engine specific impulse), and ϵ (structural factor) has been carried out. The following parameter values have been considered:

$$\sigma = 1.3, 1.4, 1.5, \tag{29a}$$

$$I_{sp} = 400, 420, 440, 460, 480, 500 \text{ s}, \tag{29b}$$

$$\epsilon = 0.08, 0.10, 0.12. \tag{29c}$$

For each $(\sigma, I_{sp}, \epsilon)$ combination, the maximum payload weight problem (25) has been solved with SGRA. For details, see Refs. 1 and 3. For a particular value of the initial thrust-to-weight ratio ($\sigma = 1.4$), the maximum value of the normalized payload weight W_*/W_i is plotted in Fig. 5 versus the specific impulse I_{sp} for the values (29c) of the structural factor.

The following comments can be made.

(i) The normalized payload weight of SSTO configurations increases as the engine specific impulse increases, but decreases as the spacecraft structural factor increases.

(ii) The design of SSTO configurations might be comfortably feasible, marginally feasible, or unfeasible, depending on the parameter values assumed. Only for $\epsilon = 0.08$ is the SSTO spacecraft feasible (payload weight positive) in the whole range of specific impulses considered.

(iii) For given values of σ and ϵ, feasibility can be achieved providing the engine specific impulse is above a threshold value. For $\sigma = 1.4$, the threshold value is $(I_{sp})_{min} = 425$ s if $\epsilon = 0.10$, $(I_{sp})_{min} = 464$ s if $\epsilon = 0.12$, and $(I_{sp})_{min} = 505$ s if $\epsilon = 0.14$ (not shown in Fig. 5). Note that the latter value of $(I_{sp})_{min}$ is well in excess of the values associated with the Space Shuttle engines, which are at most 450 s.

(iv) For given values of σ and I_{sp}, feasibility can be achieved providing the structural factor is below a threshold value. For $\sigma = 1.4$, this threshold value is $\epsilon_{max} = 0.087$ if $I_{sp} = 400$ s, $\epsilon_{max} = 0.107$ if $I_{sp} = 440$ s, and $\epsilon_{max} = 0.128$ if $I_{sp} = 480$ s.

4. TSTO Configuration

For the TSTO spacecraft, the maximum payload weight problem is studied first for initial thrust-to-weight ratio σ in the range 1.3 to 1.5, engine specific impulse I_{sp} in the range 400 to 500 s, and uniform structural factor $\hat{\epsilon} = \tilde{\epsilon}$ in the range 0.08 to 0.12, with $\hat{\epsilon}$ denoting the structural factor of Stage 1 and $\tilde{\epsilon}$ denoting the structural factor of Stage 2. Then, for a particular value of the initial thrust-to-weight ratio ($\sigma = 1.4$), the maximum payload weight problem is studied again for nonuniform structural factor by varying $\tilde{\epsilon}$, with $\tilde{\epsilon} \geq \hat{\epsilon}$, while keeping $\hat{\epsilon}$ fixed.

For the TSTO configuration, let a caret superscript denote Stage 1; let a tilde superscript denote Stage 2. The differential system (1) applies to both Stage 1 and Stage 2; however, owing to weight discontinuity due to staging, integration must be stopped at the end of Stage 1 and restarted at the

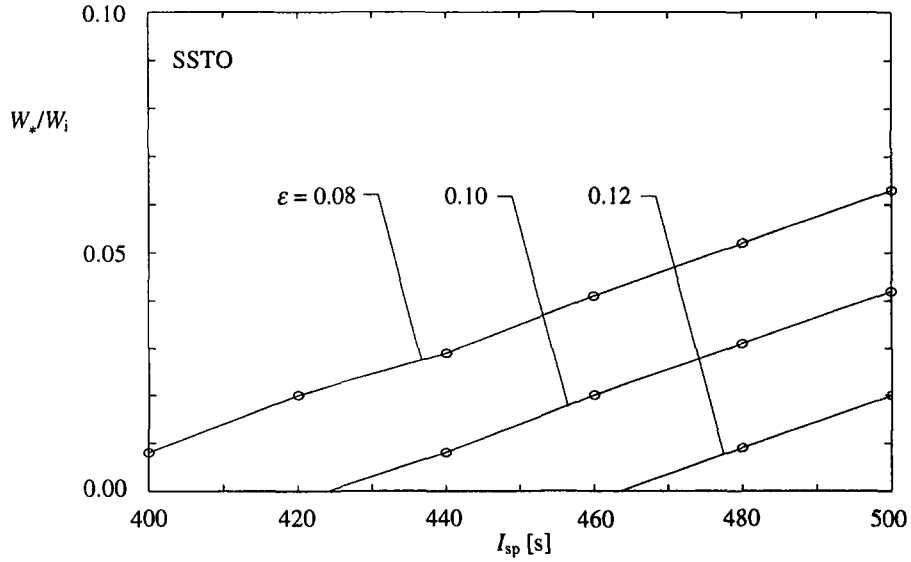

Fig. 5. SSTO spacecraft: Normalized payload weight vs specific impulse, $\sigma = 1.4$.

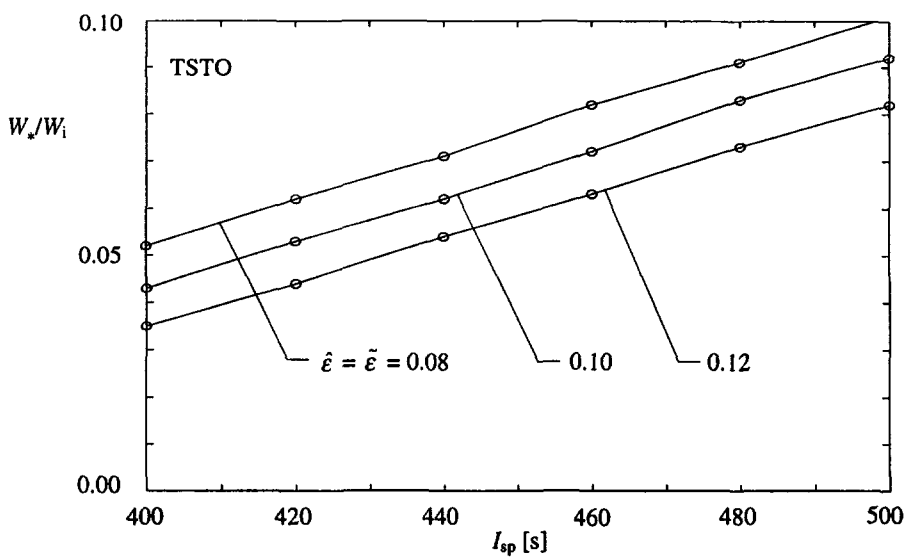

Fig. 6. TSTO spacecraft: Normalized payload weight vs specific impulse, $\sigma = 1.4$, $k = 1.0$, uniform structural factor.

beginning of Stage 2. The unknowns are the functions

$$\hat{h}(t),\ \hat{V}(t),\ \hat{\gamma}(t),\ \hat{W}(t),\ \hat{\alpha}(t),\ \hat{\beta}(t),\ \hat{\tau}, \tag{30a}$$

$$\tilde{h}(t),\ \tilde{V}(t),\ \tilde{\gamma}(t),\ \tilde{W}(t),\ \tilde{\alpha}(t),\ \tilde{\beta}(t),\ \tilde{\tau}, \tag{30b}$$

which include the state variables $\hat{h}$, $\hat{V}$, $\hat{\gamma}$, $\hat{W}$ and $\tilde{h}$, $\tilde{V}$, $\tilde{\gamma}$, $\tilde{W}$, the control variables $\hat{\alpha}$, $\hat{\beta}$ and $\tilde{\alpha}$, $\tilde{\beta}$, and the parameters $\hat{\tau}$ and $\tilde{\tau}$. The final time from takeoff to orbit is given by

$$\tau = \hat{\tau} + \tilde{\tau}. \tag{31}$$

The inequality constraints (11b) and (14b) apply to both Stage 1 and Stage 2. These are converted into equality constraints via the trigonometric transformations (12a), (15a) employed in conjunction with the differential constraints (12b), (15b). As a result, the replacements

$$\hat{\alpha}(t),\ \hat{\beta}(t) \to \hat{x}(t),\ \hat{y}(t),\ \hat{u}(t),\ \hat{w}(t), \tag{32a}$$

$$\tilde{\alpha}(t),\ \tilde{\beta}(t) \to \tilde{x}(t),\ \tilde{y}(t),\ \tilde{u}(t),\ \tilde{w}(t), \tag{32b}$$

must be introduced in (30) with $\hat{x}$, $\hat{y}$ and $\tilde{x}$, $\tilde{y}$ denoting auxiliary state variables and with $\hat{u}$, $\hat{w}$ and $\tilde{u}$, $\tilde{w}$ denoting auxiliary control variables. The path constraints (17) apply to both Stage 1 and Stage 2. At the interface between stages, there is a weight discontinuity due to staging; in turn, this induces a thrust discontinuity due to the requirement that the tangential acceleration be kept unchanged.

4.1. Boundary Conditions. Equations (19) must be understood as initial conditions for Stage 1; hence,

$$\hat{h}_i = 0 \text{ kft} = 0 \text{ km}, \tag{33a}$$

$$\hat{V}_i = 0 \text{ kft/s} = 0 \text{ km/s}, \tag{33b}$$

$$\hat{\gamma}_i = \pi/2, \tag{33c}$$

$$\hat{W}_i / W_r = 1, \tag{33d}$$

$$\hat{\alpha}_i = 0 \Rightarrow \hat{x}_i = 0, \tag{33e}$$

$$\hat{\beta}_i = 1 \Rightarrow \hat{y}_i = \pi/2. \tag{33f}$$

Equations (20) must be understood as final conditions for Stage 2; hence,

$$\tilde{h}_f = 1519 \text{ kft} = 463 \text{ km}, \tag{34a}$$

$$\tilde{V}_f = 25.04 \text{ kft/s} = 7.632 \text{ km/s}, \tag{34b}$$

$$\tilde{\gamma}_f = 0, \tag{34c}$$

$$\tilde{W}_f/W_r = \text{free}, \tag{34d}$$

$$\tilde{\alpha}_f = \text{free} \Rightarrow \tilde{x}_f = \text{free}, \tag{34e}$$

$$\tilde{\beta}_f = 0 \Rightarrow \tilde{y}_f = 3\pi/2. \tag{34f}$$

In Eqs. (33)–(34), the reference weight W_r is the same as the initial weight W_i. The velocity (34b) is the circular velocity at the altitude (34a).

4.2. Interface Conditions. At the interface between stages, the state variables h, V, γ, x behave continuously, while the state variables W, y behave discontinuously. The discontinuity in W is due to structural weight ejection due to staging. The discontinuity in y, hence the discontinuity in power setting β or thrust T, is due to the requirement that the tangential acceleration a_T be kept unchanged across the interface between stages. To sum up, the interface conditions (subscript i for Stage 2, subscript f for Stage 1) are as follows:

$$\tilde{h}_i = \hat{h}_f, \tag{35a}$$

$$\tilde{V}_i = \hat{V}_f, \tag{35b}$$

$$\tilde{\gamma}_i = \hat{\gamma}_f, \tag{35c}$$

$$\tilde{W}_i = \hat{W}_f - \hat{W}_s \Rightarrow \tilde{W}_i/W_r = (\hat{W}_f/W_r - \hat{\epsilon})/(1-\hat{\epsilon}), \tag{35d}$$

$$\tilde{\alpha}_i = \hat{\alpha}_f \Rightarrow \tilde{x}_i = \hat{x}_f, \tag{35e}$$

$$\tilde{\alpha}_{Ti} = \hat{\alpha}_{Tf} \Rightarrow (\tilde{T}_i \cos \tilde{\alpha}_i - \tilde{D}_i)/\tilde{W}_i = (\hat{T}_f \cos \hat{\alpha}_f - \hat{D}_f)/\hat{W}_f. \tag{35f}$$

Concerning (35d), see Eq. (41); concerning (35f), see Eq. (16a).

4.3. Weight Distribution. Relations (21)–(24), written for the SSTO configuration, are valid for the TSTO configuration with the following understanding: each relation must be rewritten with a caret superscript for Stage 1 and a tilde superscript for Stage 2. Concerning the structural factor, we assume that

$$\tilde{\epsilon} = k\hat{\epsilon}, \tag{36a}$$

with

$$k \geq 1. \tag{36b}$$

In particular, for Stage 1, we have [see (23)]

$$\hat{W}_p = \hat{W}_i - \hat{W}_f, \tag{37a}$$

$$\hat{W}_s = \hat{\epsilon}(\hat{W}_i - \hat{W}_f)/(1-\hat{\epsilon}), \tag{37b}$$

$$\hat{W}_* = (\hat{W}_f - \hat{\epsilon}\hat{W}_i)/(1-\hat{\epsilon}), \tag{37c}$$

with

$$\hat{W}_{\mathrm{i}} = \hat{W}_{\mathrm{p}} + \hat{W}_{\mathrm{s}} + \hat{W}_{*}. \tag{38}$$

For Stage 2, we have [see (23)]

$$\tilde{W}_{\mathrm{p}} = \tilde{W}_{\mathrm{i}} - \tilde{W}_{\mathrm{f}}, \tag{39a}$$

$$\tilde{W}_{\mathrm{s}} = \tilde{\epsilon}(\tilde{W}_{\mathrm{i}} - \tilde{W}_{\mathrm{f}})/(1-\tilde{\epsilon}), \tag{39b}$$

$$\tilde{W}_{*} = (\tilde{W}_{\mathrm{f}} - \tilde{\epsilon}\tilde{W}_{\mathrm{i}})/(1-\tilde{\epsilon}), \tag{39c}$$

with

$$\tilde{W}_{\mathrm{i}} = \tilde{W}_{\mathrm{p}} + \tilde{W}_{\mathrm{s}} + \tilde{W}_{*}. \tag{40}$$

The transition condition at the interface between stages is

$$\tilde{W}_{\mathrm{i}} = \hat{W}_{*} = \hat{W}_{\mathrm{f}} - \hat{W}_{\mathrm{s}} = (\hat{W}_{\mathrm{f}} - \hat{\epsilon}\hat{W}_{\mathrm{i}})/(1-\hat{\epsilon}). \tag{41}$$

For the TSTO configuration as a whole, the following definitions hold:

$$W_{\mathrm{i}} = \hat{W}_{\mathrm{i}}, \qquad W_{\mathrm{f}} = \tilde{W}_{\mathrm{f}}, \tag{42a}$$

$$W_{\mathrm{p}} = \hat{W}_{\mathrm{p}} + \tilde{W}_{\mathrm{p}}, \qquad W_{\mathrm{s}} = \hat{W}_{\mathrm{s}} + \tilde{W}_{\mathrm{s}}, \qquad W_{*} = \tilde{W}_{*}, \tag{42b}$$

with the implication that

$$W_{\mathrm{i}} = W_{\mathrm{p}} + W_{\mathrm{s}} + W_{*}, \tag{43}$$

as can be seen by adding (38) and (40), and then accounting for (37c), (41), (42).

4.4. Optimization Problem. For the TSTO configuration, the maximum payload weight problem can be formulated as follows:

$$\max \quad J = W_{*} = \tilde{W}_{*} = (\tilde{W}_{\mathrm{f}} - \tilde{\epsilon}\tilde{W}_{\mathrm{i}})/(1-\tilde{\epsilon}), \tag{44a}$$

$$\text{s.t.} \quad (1), (12), (15), (17), (33), (34), (35). \tag{44b}$$

The unknowns are the functions

$$\hat{h}(t),\ \hat{V}(t),\ \hat{\gamma}(t),\ \hat{W}(t),\ \hat{x}(t),\ \hat{y}(t),\ \hat{u}(t),\ \hat{w}(t),\ \hat{\tau}, \tag{45a}$$

$$\tilde{h}(t),\ \tilde{V}(t),\ \tilde{\gamma}(t),\ \tilde{W}(t),\ \tilde{x}(t),\ \tilde{y}(t),\ \tilde{u}(t),\ \tilde{w}(t),\ \tilde{\tau}, \tag{45b}$$

with the caret superscript denoting Stage 1 and tilde superscript denoting Stage 2. The unknowns (45) include the original state variables $\hat{h}$, $\hat{V}$, $\hat{\gamma}$, $\hat{W}$ and $\tilde{h}$, $\tilde{V}$, $\tilde{\gamma}$, $\tilde{W}$, the auxiliary state variables $\hat{x}$, $\hat{y}$ and $\tilde{x}$, $\tilde{y}$, the auxiliary control variables $\hat{u}$, $\hat{w}$ and $\tilde{u}$, $\tilde{w}$, and the parameters $\hat{\tau}$ and $\tilde{\tau}$. With the solution known, the original control variables $\hat{\alpha}$, $\hat{\beta}$ and $\tilde{\alpha}$, $\tilde{\beta}$, can be recovered via

(12a) and (15a). Also, the total time from takeoff to orbit is

$$\tau = \hat{\tau} + \tilde{\tau}. \tag{46}$$

The following comments are pertinent.

(i) Concerning the path constraints (17), the nature of the problem is such that (17b) and (17c) are satisfied always with strict inequality. Therefore, only (17a) is accounted for via a penalty functional added to the functional being minimized.

(ii) In treating problem (44), it is convenient to normalize the dimensional time t via the transformations

$$\theta = t/\hat{\tau}, \qquad 0 \le t \le \hat{\tau}, \tag{47a}$$

$$\theta = 1 + (t - \hat{\tau})/\tilde{\tau}, \qquad \hat{\tau} \le t \le \hat{\tau} + \tilde{\tau} = \tau, \tag{47b}$$

where θ is the dimensionless time, $\hat{\tau}$ the time length of Stage 1, $\tilde{\tau}$ the time length of Stage 2, and τ the final time. Clearly,

$$0 \le \theta \le 1, \qquad \text{if } 0 \le t \le \hat{\tau}, \tag{48a}$$

$$1 \le \theta \le 2, \qquad \text{if } \hat{\tau} \le t \le \hat{\tau} + \tilde{\tau} = \tau. \tag{48b}$$

Therefore, the dimensionless time has the value $\theta = 0$ at the beginning of Stage 1, $\theta = 1$ at the interface between stages, and $\theta = 2$ at the end of Stage 2.

(iii) The singularity present at the initial point, due to the fact that $\hat{V}_i = 0$, is handled in the same way as in Section 3.3.

(iv) The approach taken to solve this problem is the same as that described in Section 3.3, with the additional complication that the switch time $\hat{\tau}$ from Stage 1 to Stage 2 must be determined in addition to the final time τ. A further complication is the presence of the interface conditions (35) in addition to the initial conditions (33) and final conditions (34). With reference to (35), note that (35a), (35b), (35c), (35e) are continuity conditions, while (35d) and (35f) are discontinuity conditions: (35d) yields a discontinuity in weight consistent with the fact that structural weight is being ejected due to staging; (35f) yields a discontinuity in thrust consistent with the requirement that the tangential acceleration be continuous across the interface between stages.

(v) The optimization problem under consideration is of the Bolza type and can be solved via the sequential gradient-restoration algorithm for optimal control problems (SGRA, Refs. 8–10). For the TSTO configuration, since the payload weight depends not only on final conditions but also on interface (staging) conditions, modifications of the original SGRA formulation have been necessary; further modifications have been needed to deal with the discontinuities present in this case.

4.5. Results: Uniform Structural Factor. In this study, the following parameter values have been considered:

$$\sigma = 1.3,\ 1.4,\ 1.5, \tag{49a}$$

$$I_{sp} = 400,\ 420,\ 440,\ 460,\ 480,\ 500\ \text{s}, \tag{49b}$$

$$\hat{\epsilon} = \tilde{\epsilon} = 0.08,\ 0.10,\ 0.12. \tag{49c}$$

For each (σ, I_{sp}, $\hat{\epsilon} = \tilde{\epsilon}$) combination, the maximum payload weight problem (44) has been solved with SGRA. For details, see Refs. 1–3. For a particular value of the initial thrust-to-weight ratio ($\sigma = 1.4$), the maximum value of the normalized payload weight W_*/W_i is plotted in Fig. 6 versus the specific impulse I_{sp} for the values (49c) of the spacecraft structural factor. The following comments can be made.

(i) The normalized payload weight of TSTO configurations increases as the engine specific impulse increases, but decreases as the spacecraft structural factor increases.

(ii) The design of TSTO configurations is feasible for all of the (σ, I_{sp}, $\hat{\epsilon} = \tilde{\epsilon}$) combinations considered.

(iii) For $\sigma = 1.4$ and uniform structural factor $\hat{\epsilon} = \tilde{\epsilon}$ in the range 0.14 to 0.18 (not shown in Fig. 6), feasibility of the TSTO spacecraft can be achieved even with present-day engines having specific impulse around 450 s.

(iv) From the payload viewpoint, the most striking impression that one gets from Figs. 5–6 is the clear superiority of the TSTO configuration vis-a-vis the SSTO configuration; the normalized payload gap between the two configurations increases as the structural factor increases.

(v) To sum up, for the same structural factor, the normalized payload of a TSTO configuration is considerably larger than that of a SSTO configuration. Conversely, for the same normalized payload, one can design a TSTO configuration having a much larger structural factor than the SSTO configuration, with the implication of increased safety and reliability.

4.6. Results: Nonuniform Structural Factor. In this study, the following parameter values have been considered:

$$\sigma = 1.4, \tag{50a}$$

$$I_{sp} = 400,\ 420,\ 440,\ 460,\ 480,\ 500\ \text{s}, \tag{50b}$$

$$\hat{\epsilon} = 0.08,\ 0.10,\ 0.12, \tag{50c}$$

$$\tilde{\epsilon} = k\hat{\epsilon}, \qquad k = 1.0,\ 1.5,\ 2.0,\ 2.5,\ 3.0. \tag{50d}$$

For each $(\sigma, I_{sp}, \hat{\epsilon}, \tilde{\epsilon} = k\hat{\epsilon})$ combination, the maximum payload weight problem (44) has been solved with SGRA. For details, see Refs. 1–3. The normalized payload weight W_*/W_i is plotted in Figs. 6–10 versus the engine specific impulse for three values of the Stage 1 structural factor and the values (50d) of the parameter k, which is the ratio of the structural factors of Stage 2 and Stage 1. The following comments can be made.

(i) The normalized payload weight increases as the engine specific impulse increases, but decreases as the Stage 1 structural factor increases and as the parameter k increases, hence as the Stage 2 structural factor increases.

(ii) Even if the Stage 2 structural factor is twice the Stage 1 structural factor, the TSTO configuration is feasible.

(iii) For $I_{sp} = 440$ s, the maximum value of the parameter $k = \tilde{\epsilon}/\hat{\epsilon}$ for which feasibility (payload weight positive) can be guaranteed is $k = 3.5$ for $\hat{\epsilon} = 0.08$, $k = 2.8$ for $\hat{\epsilon} = 0.10$, and $k = 2.3$ for $\hat{\epsilon} = 0.12$; this corresponds to a Stage 2 structural factor $\tilde{\epsilon} \cong 0.28$, nearly independent of the Stage 1 structural factor $\hat{\epsilon}$.

5. Effect of Drag Changes

After completing the previous optimization studies on SSTO and TSTO configurations (Sections 3–4), an additional topic was considered, namely, the effect of drag changes on payload weight. Suppose that the drag and lift of the spacecraft are embedded into a one-parameter family of the form

$$D' = \eta D, \qquad L' = L, \tag{51}$$

where η is the drag factor. Clearly, $\eta = 1$ yields the drag and lift of the baseline configuration; $\eta = 0.5$ reduces the drag by 50%, while keeping the lift unchanged; $\eta = 1.5$ increases the drag by 50%, while keeping the lift unchanged.

The maximum payload weight problem was solved again for $\eta = 0.5$, $\eta = 1.0$, and $\eta = 1.5$. The following parameter values were considered:

$$\text{(SSTO)} \qquad \sigma = 1.4 \qquad I_{sp} = 440 \text{ s}, \qquad \epsilon = 0.10, \tag{52}$$

$$\text{(TSTO)} \qquad \sigma = 1.4, \qquad I_{sp} = 440 \text{ s}, \qquad \hat{\epsilon} = \tilde{\epsilon} = 0.10, \tag{53}$$

with (53) indicating that a uniform structural factor is being considered for the TSTO configuration. The results are shown in Figs. 11–12, where the normalized payload weight is plotted versus the drag factor $\eta = D'/D$ for the parameters choices (52)–(53).

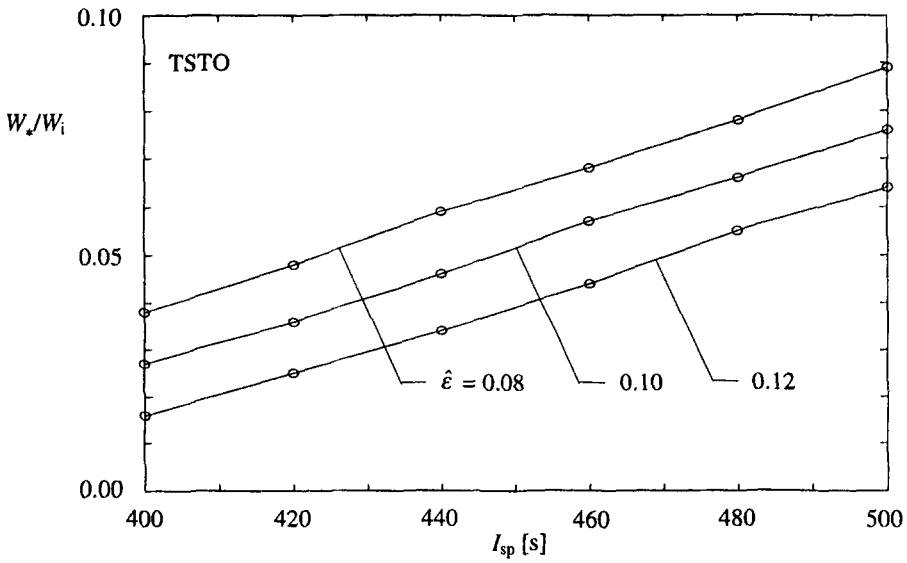

Fig. 7. TSTO spacecraft: Normalized payload weight vs specific impulse, $\sigma = 1.4$, $k = 1.5$, nonuniform structural factor.

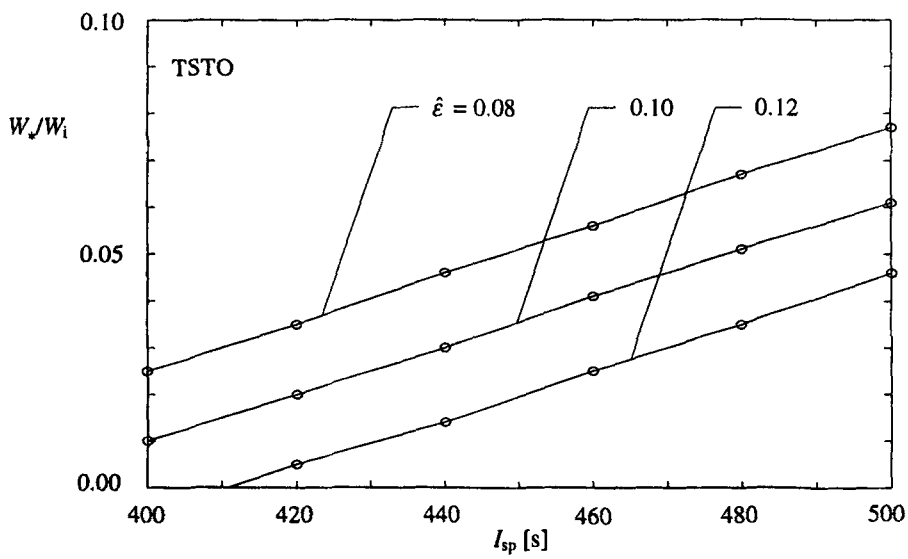

Fig. 8. TSTO spacecraft: Normalized payload weight vs specific impulse, $\sigma = 1.4$, $k = 2.0$, nonuniform structural factor.

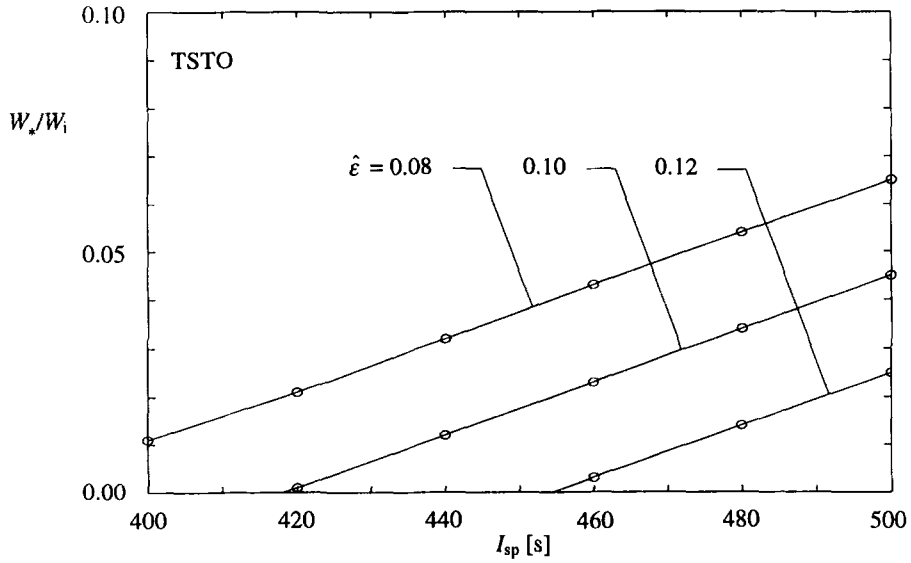

Fig. 9. TSTO spacecraft: Normalized payload weight vs specific impulse, $\sigma = 1.4$, $k = 2.5$, nonuniform structural factor.

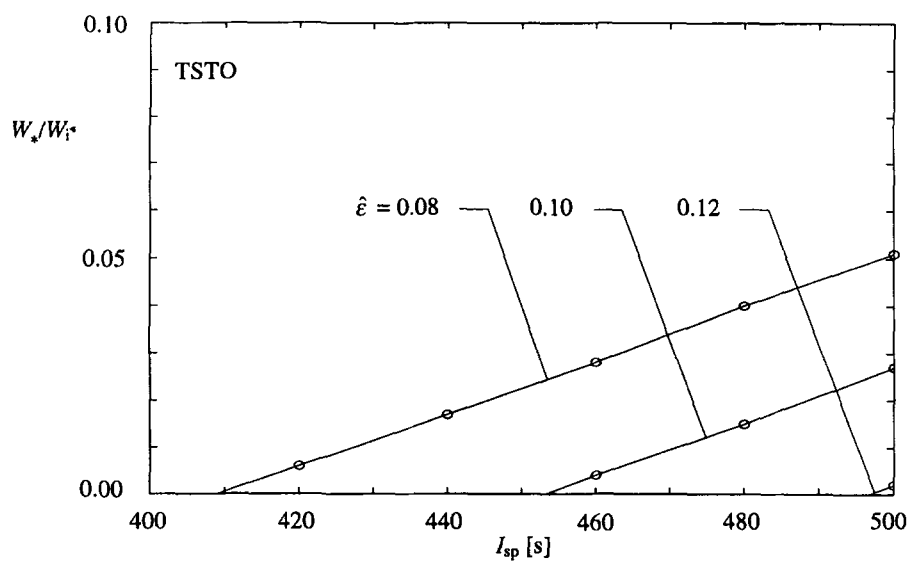

Fig. 10. TSTO spacecraft: Normalized payload weight vs specific impulse, $\sigma = 1.4$, $k = 3.0$, nonuniform structural factor.

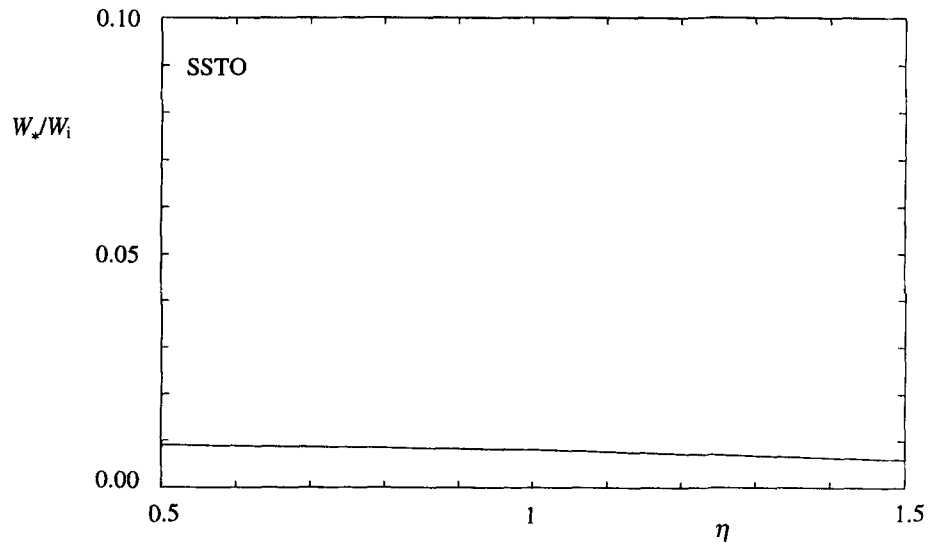

Fig. 11. SSTO spacecraft: Normalized payload weight vs drag factor $\eta = D'/D$, $\sigma = 1.4$, $I_{sp} = 440$ s, $\epsilon = 0.10$.

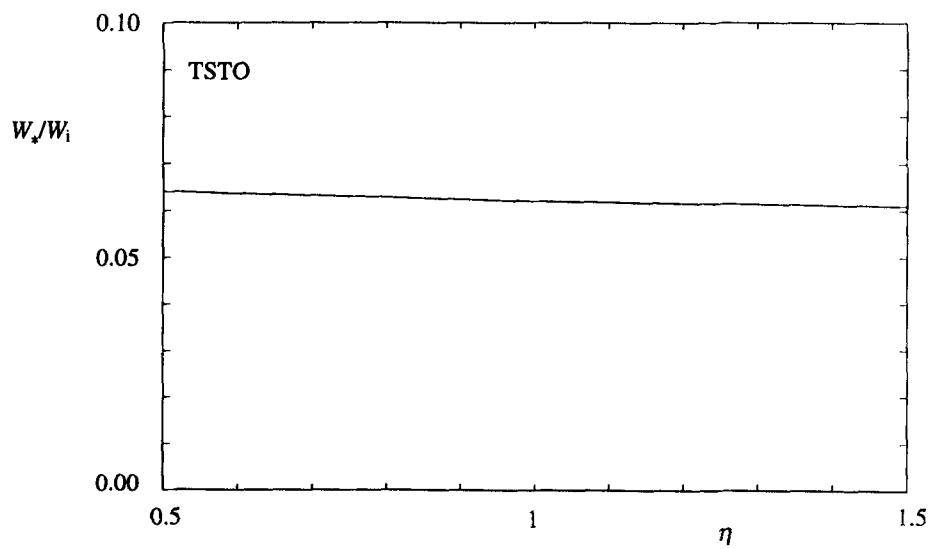

Fig. 12. TSTO spacecraft: Normalized payload weight vs drag factor $\eta = D'/D$, $\sigma = 1.4$, $I_{sp} = 440$ s, $\hat{\epsilon} = \tilde{\epsilon} = 0.10$.

Consistently with the results of Refs. 7 and 16–18, the analysis shows that changing the drag by ±50% produces relatively small changes in final weight and payload weight. One surmises that the final weight and payload weight are not very sensitive to the aerodynamic model of the spacecraft, or equivalently that the aerodynamic forces do not have a large influence on propellant consumption. Indeed, should an energy balance be made, one would find that the largest part of the energy produced by the rocket powerplant is spent in accelerating the spacecraft to the final velocity; only a minor part is spent in overcoming aerodynamic and gravitational effects.

For TSTO configurations, these results justify having neglected in the analysis drag changes due to staging, and hence having assumed that the drag function of Stage 2 is the same as the drag function of Stage 1.

6. Design Considerations

The results of Section 3 show that, for SSTO configurations, the normalized payload weight W_*/W_i behaves almost linearly with respect to the engine specific impulse I_{sp} and the spacecraft structural factor ϵ (Fig. 5). The results of Section 4 show that the same behavior is true for TSTO configurations as long as the ratio k of the structural factors of Stage 2 and Stage 1 is held constant (Figs. 6–10). This leads to useful design considerations.

6.1. SSTO Configuration. The assumed linear representation of the results of Section 3 is the following:

$$W_*/W_i \cong A + B\epsilon + CI_{sp}/I_{spr}, \tag{54}$$

where A, B, C are dimensionless constants and $I_{spr} = 1000$ s is a reference specific impulse. The representation (54) implies that the zero-payload condition ($W_* = 0$) separating feasibility ($W_* > 0$) from unfeasibility ($W_* < 0$) is given by

$$\epsilon \cong D + EI_{sp}/I_{spr}, \tag{55}$$

with

$$D = -A/B, \qquad E = -C/B. \tag{56}$$

The constants in Eq. (54) can be obtained via a least-square fit of the computed normalized payload weights W_*/W_i, treated as available data in the least-square process,

$$A = -0.1282, \qquad B = -1.0958, \qquad C = 0.5590, \tag{57}$$

implying that the constants in Eq. (55) are given by

$$D=-0.1170, \qquad E=0.5101. \tag{58}$$

The normalized payload weight predicted via (54) and (57) is in excellent agreement with the available data, shown as circles in Fig. 5. Consistently, the zero-payload condition predicted via (55) and (58) is shown in Fig. 13 with reference to the (I_{sp}, ϵ) plane. In Fig. 13, the limiting line $W_*=0$ is nonconservative, since part of the payload weight has to be used as propellant for space maneuvers, reentry and landing maneuvers, and reserve margin for emergency. Hence, an actual design must lie wholly inside the feasible region of Fig. 13.

6.2. TSTO Configuration: Uniform Structural Factor. The representation (54)–(55) can be extended to a TSTO configuration with uniform structural factor if one sets

$$\epsilon=\hat{\epsilon}=\tilde{\epsilon}. \tag{59}$$

In this case, the constants (57) must be replaced by

$$A=-0.1059, \qquad B=-0.4500, \qquad C=0.4852, \tag{60}$$

with the implication that

$$D=-0.2353, \qquad E=1.0782. \tag{61}$$

The normalized payload weight predicted via (54) and (60) is in excellent agreement with the available data, shown as circles in Fig. 6. Consistently, the zero-payload condition predicted via (55) and (61) is shown in Fig. 14 with reference to the (I_{sp}, ϵ) plane. In Fig. 14, the limiting line $W_*=0$ is nonconservative, since part of the payload weight has to be used as propellant for space maneuvers, reentry and landing maneuvers, and reserve margin for emergency. Hence, an actual design must lie wholly inside the feasible region of Fig. 14.

6.3. TSTO Configuration: Nonuniform Structural Factor. The representation (54)–(55) can be further extended to a TSTO configuration with nonuniform structural factor if one sets

$$\hat{\epsilon}=\epsilon, \qquad \tilde{\epsilon}=k\epsilon, \tag{62}$$

where k is a parameter. In this case, the constants in Eq. (54) must be replaced by functions of the parameter k, so that the representation becomes

$$W_*/W_i \cong A(k)+B(k)\epsilon+C(k)I_{sp}/I_{spr}, \tag{63}$$

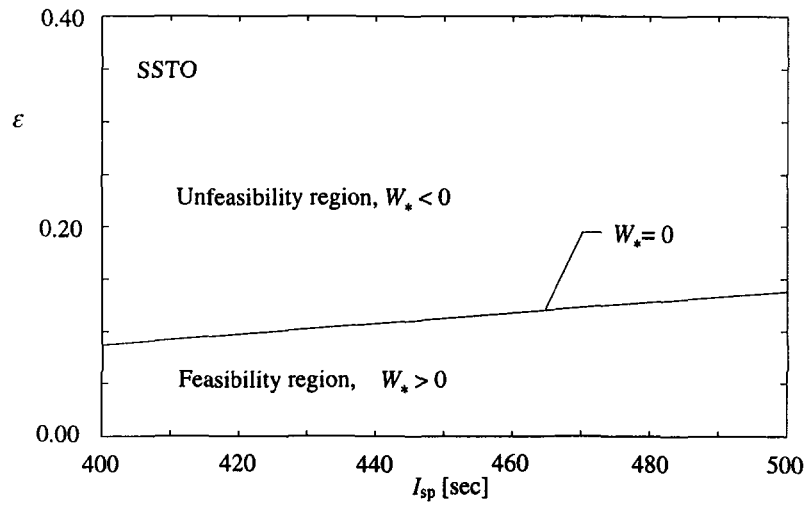

Fig. 13. SSTO spacecraft: Structural factor vs specific impulse, $\sigma = 1.4$, zero-payload line.

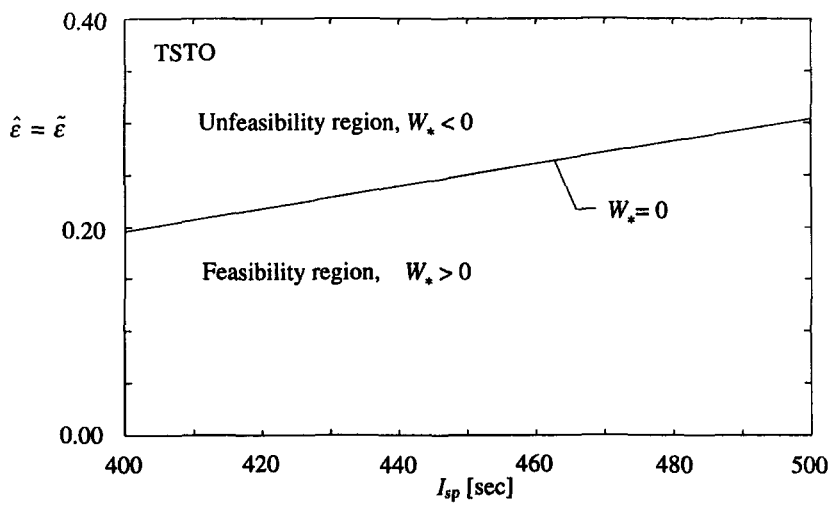

Fig. 14. TSTO spacecraft: Structural factor vs specific impulse, $\sigma = 1.4$, $k = 1.0$, uniform structural factor, zero-payload line.

with the implication that the zero-payload condition ($W_* = 0$) separating feasibility ($W_* > 0$) from unfeasibility ($W_* < 0$) is given by

$$\epsilon \cong D(k) + E(k) I_{sp}/I_{spr}, \tag{64}$$

with

$$D(k) = -A(k)/B(k), \qquad E(k) = -C(k)/B(k). \tag{65}$$

Five values of the parameter k are considered, namely,

$$k = 1.0,\ 1.5,\ 2.0,\ 2.5,\ 3.0. \tag{66}$$

Therefore, the constants (60)–(61) must be replaced by

$$A(1.0) = -0.1059, \qquad B(1.0) = -0.4500, \qquad C(1.0) = 0.4852, \tag{67a}$$

$$A(1.5) = -0.1122, \qquad B(1.5) = -0.5917, \qquad C(1.5) = 0.4952, \tag{67b}$$

$$A(2.0) = -0.1193, \qquad B(2.0) = -0.7792, \qquad C(2.0) = 0.5171, \tag{67c}$$

$$A(2.5) = -0.1266, \qquad B(2.5) = -1.0000, \qquad C(2.5) = 0.5429, \tag{67d}$$

$$A(3.0) = -0.1325, \qquad B(3.0) = -1.2542, \qquad C(3.0) = 0.5686, \tag{67e}$$

and

$$D(1.0) = -0.2353, \qquad E(1.0) = 1.0782, \tag{68a}$$

$$D(1.5) = -0.1896, \qquad E(1.5) = 0.8369, \tag{68b}$$

$$D(2.0) = -0.1531, \qquad E(2.0) = 0.6636, \tag{68c}$$

$$D(2.5) = -0.1266, \qquad E(2.5) = 0.5429, \tag{68d}$$

$$D(3.0) = -0.1056, \qquad E(3.0) = 0.4534. \tag{69e}$$

In turn, D and E behave almost quadratically with respect to the parameter k. Hence, the assumed representation is the following:

$$D(k) = D_0 + D_1 k + D_2 k^2, \tag{69a}$$

$$E(k) = E_0 + E_1 k + E_2 k^2. \tag{69b}$$

A least-square fit of the available data (68) with the quadratic functions (69) produces the following constants:

$$D_0 = -0.3501, \qquad D_1 = +0.1320, \qquad D_2 = -0.0169, \tag{70a}$$

$$E_0 = +1.6886, \qquad E_1 = -0.7156, \qquad E_2 = +0.1017. \tag{70b}$$

For each of the values (66) of the parameter k, the normalized payload weight predicted via (63) and (67) is in excellent agreement with the available data, shown as circles in Figs. 6–10. Consistently, for each of the values (66)

of the parameter k, the zero-payload condition predicted via (64) and (68) is shown in Figs. 14–18 with reference to the (I_{sp}, ϵ) plane. In Figs. 14–18, the limiting line $W_* = 0$ is nonconservative, since part of the payload weight has to be used as propellant for space maneuvers, reentry and landing maneuvers, and reserve margin for emergency. Hence, the actual design must lie wholly inside the feasible region of Figs. 14–18.

As a final observation, we note that, for given $\epsilon = \hat{\epsilon}$, the zero-payload lines $W_* = 0$ shift downward as the parameter k increases, thereby shrinking the size of the feasible region as k increases.

7. Summary and Conclusions

In this paper, the maximum payload weight problem has been solved for both a single-stage-to-orbit (SSTO) and a two-stage-to-orbit (TSTO) rocket-powered spacecraft. Optimal ascent trajectories have been computed, first for SSTO spacecraft with initial thrust-to-weight ratio σ in the range 1.3 to 1.5, engine specific impulse I_{sp} in the range 400 to 500 s, and structural factor ϵ in the range 0.08 to 0.12. Then, the study has been extended to TSTO spacecraft with uniform structural factor $\hat{\epsilon} = \tilde{\epsilon}$. Finally, the study has been extended to TSTO spacecraft with nonuniform structural factor $\tilde{\epsilon} = k\hat{\epsilon}$, with $k \geq 1$. The main conclusions are as follows.

(i) Feasibility of SSTO configurations can be achieved with only part of the parameter combinations considered. Feasibility of TSTO configurations can be achieved with all the parameter combinations considered.

(ii) For the case of uniform structural factor, comparison of the SSTO and TSTO configurations for the same values of σ, I_{sp}, and the same structural factor shows that the maximum payload of the latter is several times that of the former.

(iii) For the case of nonuniform structural factor, comparison of the SSTO and TSTO configurations for $\sigma = 1.4$, $I_{sp} = 440$ s, and structural factors $\hat{\epsilon} = \epsilon$ and $\tilde{\epsilon} = 1.5\,\epsilon$ shows that the TSTO payload is about double the SSTO payload for $\epsilon = 0.08$ and more than quintuple the TSTO payload for $\epsilon = 0.10$.

(iv) For the case of nonuniform structural factor, comparison of the SSTO and TSTO configurations for the same values of σ, I_{sp}, and the same payload shows that, if $\epsilon = \hat{\epsilon}$, then one can design a TSTO configuration with $\tilde{\epsilon}$ considerably larger than $\hat{\epsilon}$, implying increased safety and reliability of the TSTO configuration vis-a-vis the SSTO configuration. The fact that $\tilde{\epsilon}$ can be made larger than $\hat{\epsilon}$ suggests that an attractive TSTO design might be a first-stage structure made of only tanks and a second-stage structure made of engines, tanks, electronics, and so on.

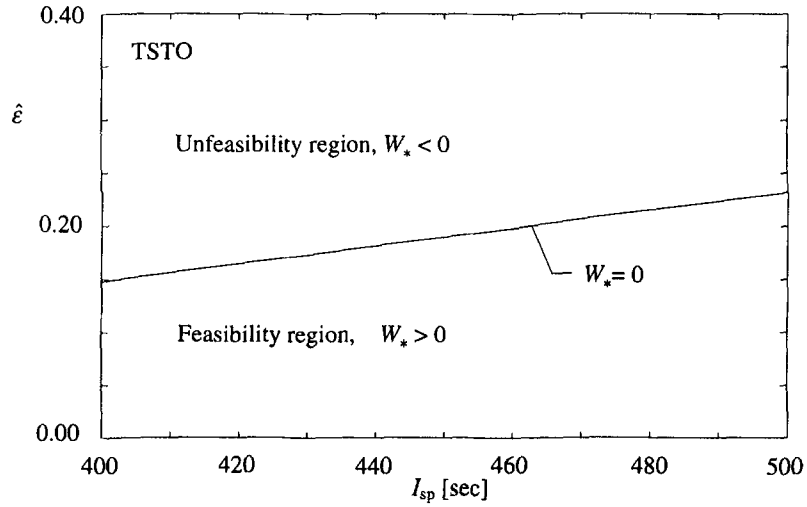

Fig. 15. TSTO spacecraft: Stage 1 structural factor vs specific impulse, $\sigma = 1.4$, $k = 1.5$, nonuniform structural factor, zero-payload line.

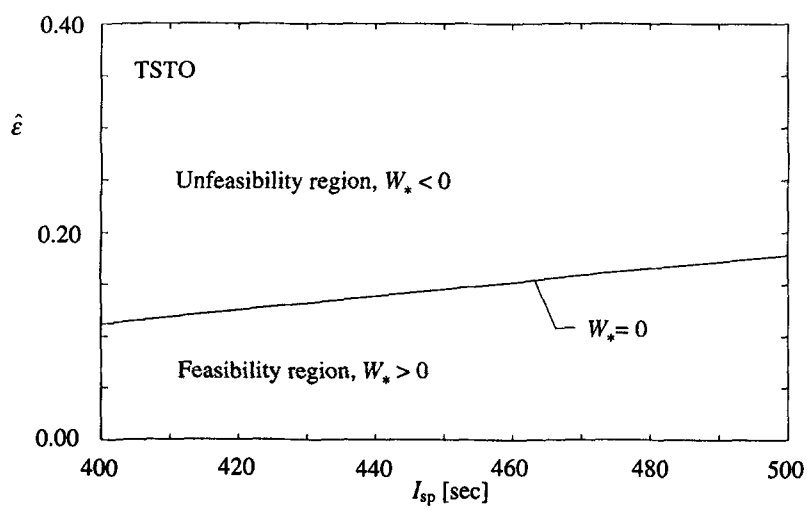

Fig. 16. TSTO spacecraft: Stage 1 structural factor vs specific impulse, $\sigma = 1.4$, $k = 2.0$, nonuniform structural factor, zero-payload line.

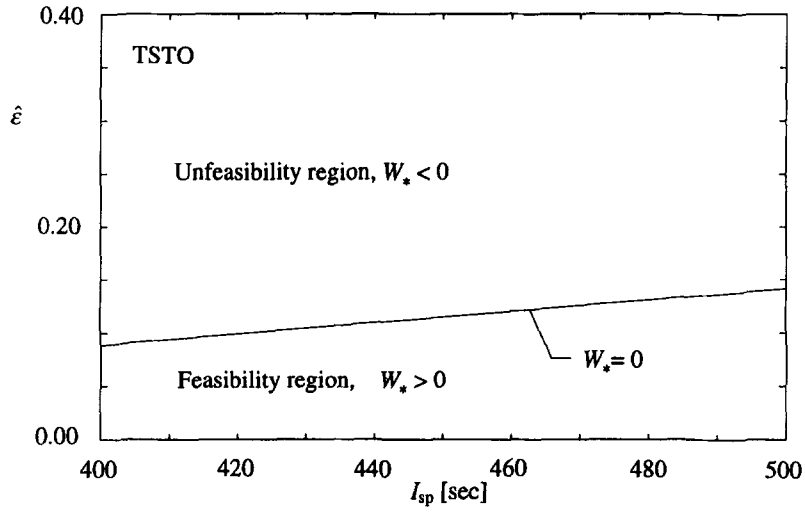

Fig. 17. TSTO spacecraft: Stage 1 structural factor vs specific impulse, $\sigma = 1.4$, $k = 2.5$, nonuniform structural factor, zero-payload line.

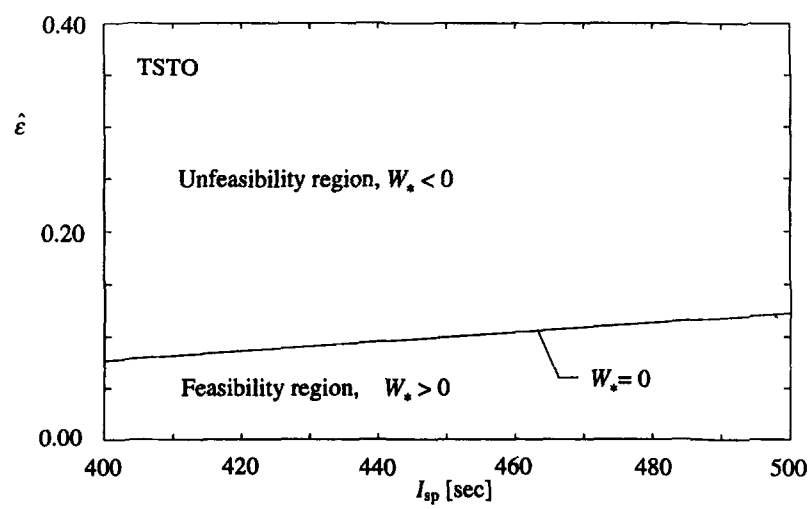

Fig. 18. TSTO spacecraft: Stage 1 structural factor vs specific impulse, $\sigma = 1.4$, $k = 3.0$, nonuniform structural factor, zero-payload line.

(v) For SSTO feasibility to became reality, improvements in the engine specific impulse and spacecraft structural factor are desirable, indeed crucial. Paradoxically, aerodynamic improvements are less important in that they yield relatively small improvements in payload weight.

(vi) The maximum payload weight behaves almost linearly with respect to the engine specific impulse and the spacecraft structural factor. This property, first established for SSTO configurations, holds for TSTO configurations as long as the ratio of the Stage 2 structural factor to Stage 1 structural factor is held constant. With reference to the specific impulse/structural factor domain, this property leads to the construction of a zero-payload line separating the feasibility region (positive payload) from the unfeasibility region (negative payload).

(vii) Although the design of a SSTO spacecraft is attractive from a practical point of view (complete reusability of the spacecraft), its feasibility appears to be strongly dependent on future improvements in the areas of engine specific impulse and spacecraft structural factor. Therefore, in approaching the design of next-generation spacecraft, prudence suggests that a TSTO spacecraft be given concurrent consideration, especially if it is not possible to achieve major improvements in engine specific impulse.

References

1. MIELE, A., and MANCUSO, S., *Optimal Ascent Trajectories for SSTO and TSTO Spacecraft*, Aero-Astronautics Report 276, Rice University, 1997.
2. MIELE, A., and MANCUSO, S., *Optimal Ascent Trajectories for TSTO Spacecraft: Extensions*, Aero-Astronautics Report 277, Rice University, 1997.
3. MIELE, A., and MANCUSO, S., *Optimal Ascent Trajectories for SSSO, SSTO, and TSTO Spacecraft: Extensions*, Aero-Astronautics Report 278, Rice University, 1997.
4. ANONYMOUS, N. N., *Access to Space Study, Summary Report*, Office of Space Systems Development, NASA Headquarters, 1994.
5. FREEMAN, D. C., TALAY, T. A., STANLEY, D. O., LEPSCH, R. A., and WIHITE, A. W., *Design Options for Advanced Manned Launch Systems*, Journal of Spacecraft and Rockets, Vol. 32, No. 2, pp. 241–249, 1995.
6. GREGORY, I. M., CHOWDHRY, R. S., and MCMIMM, J. D., *Hypersonic Vehicle Model and Control Law Development Using H_∞ and μ-Synthesis*, Technical Memorandum 4562, NASA, 1994.
7. MIELE, A., and MANCUSO, S., *Optimal Ascent Trajectories for a Single-Stage Suborbital Spacecraft*, Aero-Astronautics Report 275, Rice University, 1997.
8. MIELE, A., WANG, T., and BASAPUR, V. K., *Primal and Dual Formulations of Sequential Gradient-Restoration Algorithms for Trajectory Optimization Problems*, Acta Astronautica, Vol. 13, No. 8, pp. 491–505, 1986.

9. MIELE, A., and WANG, T., *Primal-Dual Properties of Sequential Gradient-Restoration Algorithms for Optimal Control Problems, Part 1: Basic Problem*, Integral Methods in Science and Engineering, Edited by F. R. Payne et al, Hemisphere Publishing Corporation, Washington, DC, pp. 577–607, 1986.
10. MIELE, A., and WANG, T., *Primal-Dual Properties of Sequential Gradient-Restoration Algorithms for Optimal Control Problems, Part 2: General Problem*, Journal of Mathematical Analysis and Applications, Vol. 119, Nos. 1–2, pp. 21–54, 1986.
11. RISHIKOF, B. H., MCCORMICK, B. R., PRITCHARD, R. E., and SPONAUGLE, S. J., *SEGRAM: A Practical and Versatile Tool for Spacecraft Trajectory Otpimization*, Acta Astronautica, Vol. 26, Nos. 8–10, pp. 599–609, 1992.
12. MIELE, A., and WANG, T., *Optimization and Acceleration Guidance of Flight Trajectories in a Windshear*, Journal of Guidance, Control, and Dynamics, Vol. 10, No. 4, pp. 368–377, 1987.
13. MIELE, A., and WANG, T., *Acceleration, Gamma, and Theta Guidance for Abort Landing in a Windshear*, Journal of Guidance, Control, and Dynamics, Vol. 12, No. 6, pp. 815–821, 1989.
14. MIELE, A., and WANG, T., *Penetration Landing Guidance Trajectories in the Presence of Windshear*, Journal of Guidance, Control, and Dynamics, Vol. 12, No. 6, pp. 806–814, 1989.
15. MIELE, A., and WANG, T., *Gamma Guidance Schemes for Flight in a Windshear*, Journal of Guidance, Control, and Dynamics, Vol. 11, No. 4, pp. 320–327, 1988.
16. MIELE, A., LEE, W. Y., and WU, G. D., *Optimal Trajectories for an Aerospace Plane, Part 1: Formulation, Results, and Analysis*, Aero-Astronautics Report 247, Rice University, 1990.
17. MIELE, A., LEE, W. Y., and WU, G. D., *Optimal Trajectories for an Aerospace Plane, Part 2: Data, Tables and Graphs*, Aero-Astronautics Report 248, Rice University, 1990.
18. MIELE, A., LEE, W. Y., and WU, G. D., *Ascent Performance Feasibility of the National Aerospace Plane*, Atti della Accademia delle Scienze di Torino, Vol. 131, pp. 91–108, 1997.
19. MIELE, A., *Recent Advances in the Optimization and Guidance of Aeroassisted Orbital Transfers*, The 1st John V. Breakwell Memorial Lecture, Acta Astronautica, Vol. 38, No. 10, pp. 747–768, 1996.
20. MIELE, A., and WANG, T., *Nominal Trajectories for the Aeroassisted Flight Experiment*, Journal of the Astronautical Sciences, Vol. 41, No. 2, pp. 139–163, 1993.
21. MIELE, A., and WANG, T., *Near-Optimal Highly Robust Guidance for Aeroassisted Orbital Transfer*, Journal of Guidance, Control, and Dynamics, Vol. 19, No. 3, pp. 549–556, 1996.
22. MIELE, A., and WANG, T., *Robust Predictor-Corrector Guidance for Aeroassisted Orbital Transfer*, Journal of Guidance, Control, and Dynamics, Vol. 19, No. 5, pp. 1134–1141, 1996.
23. MIELE, A., Editor, *Special Issue on Hypervelocity Flight*, Journal of the Astronautical Sciences, Vol. 36, Nos. 1–2, pp. 1–197, 1988.

24. MIELE, A., and SALVETTI, A., Editors, *Applied Mathematics in Aerospace Sciences and Engineering*, Plenum Publishing Corporation, New York, New York, 1994.
25. MIELE, A., *Optimal Trajectories of Aircraft and Spacecraft*, Aircraft Trajectories: Computation, Prediction, and Control, Edited by A. Benoit, Agardograph AG-301, AGARD/NATO, Paris, France, Vol. 1, Chapter 2, pp. 1–56, 1990.
26. MIELE, A., *Flight Mechanics, Vol. 1: Theory of Flight Paths*, Chapters 13 and 14, Addison-Wesley Publishing Company, Reading, Massachusetts, 1962.
27. NOAA, NASA, and USAF, *US Standard Atmosphere, 1976*, US Government Printing Office, Washington, DC, 1976.